

US greenhouse gas emissions

Facility emissions reported to the EPA (GHGRP), in AR5 CO₂e. Year filter drives the cards and the segment chart.

Year

All



Total GHG emissions

2,597 Mt CO₂e

▼ -4% vs 2022



Methane

204 Mt CO₂e

▼ -2% vs 2022



F-gases

8.6 Mt CO₂e

▼ -31% vs 2022



Methane share of all GHG

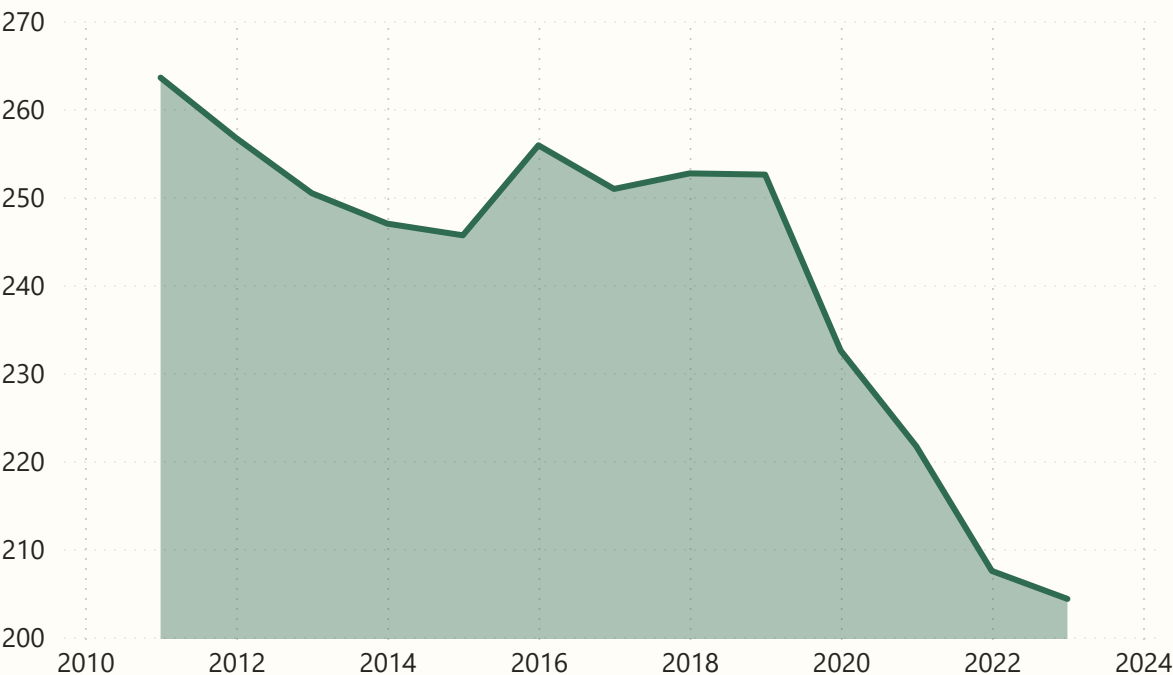
7.9%

▲ +0.2 pts vs 2022



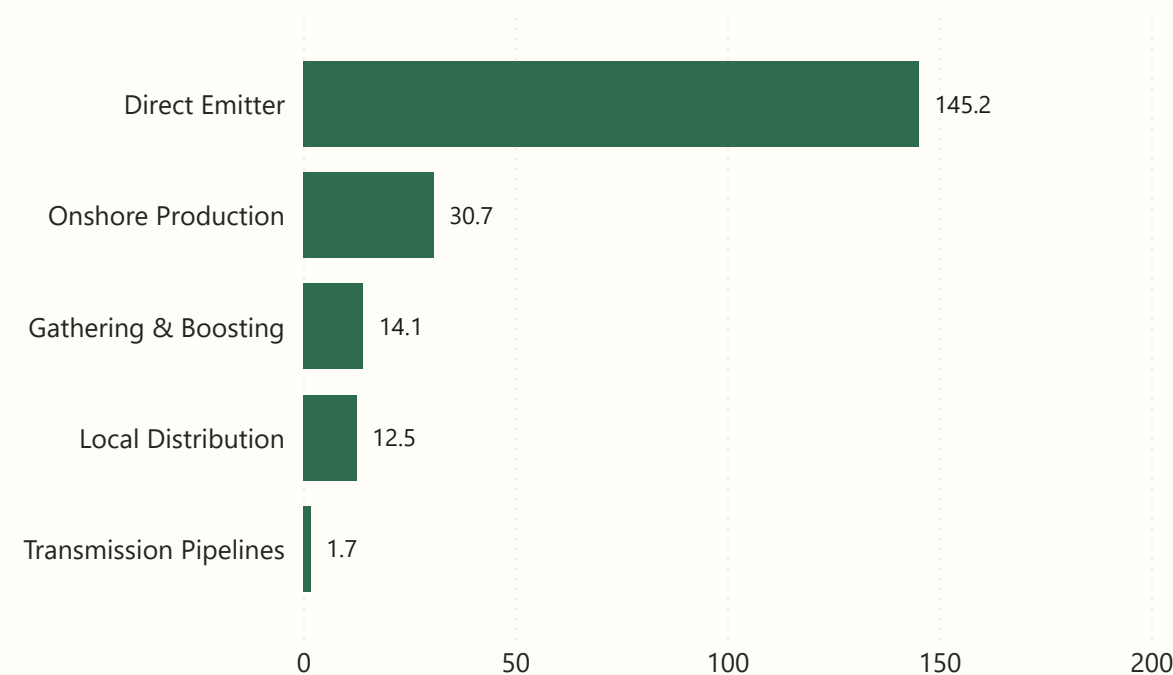
Methane emissions by year (Mt CO₂e)

Reported methane fell 22% from 2011 to 2023



Methane by reporting segment, selected year (Mt CO₂e)

Onshore production is the largest oil and gas segment



US methane outlook to 2030

Linear trend fitted to 2016-2023 EPA methane, with a 90% range and the Global Methane Pledge path. Year filter: 2024-2030.

Forecast year

All



Methane forecast

148 Mt CO₂e

90% range 120 to 177 Mt



Global Methane Pledge path

163 Mt CO₂e

-30% vs 2020 by 2030



Forecast vs pledge path

-14 Mt CO₂e

Below the pledge path (on track)



Trend per year

-8.2 Mt CO₂e/yr

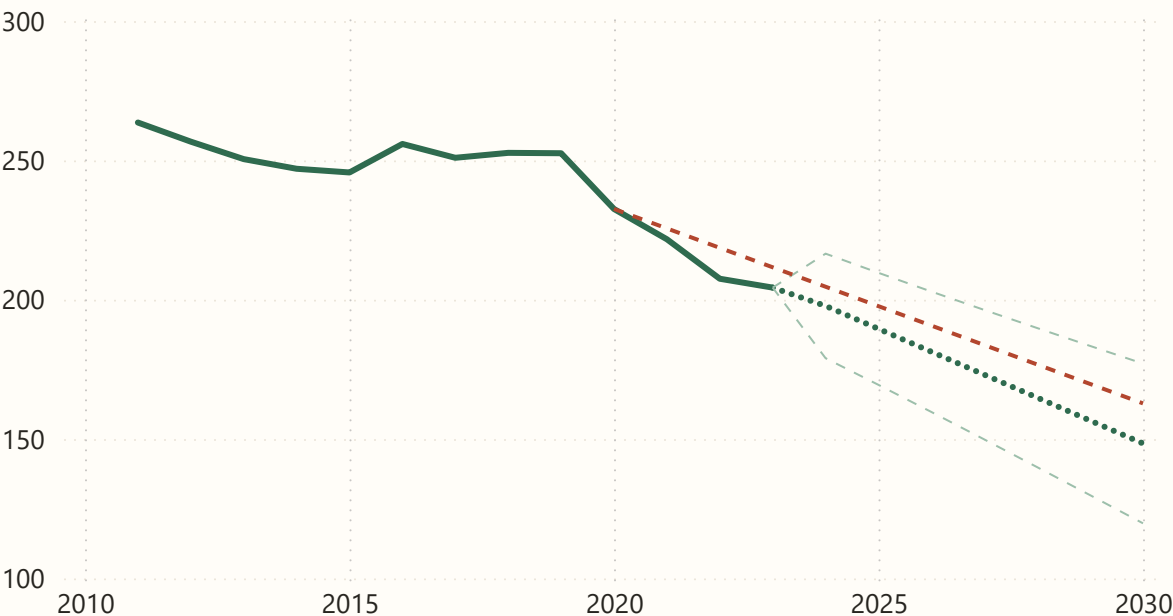
Linear fit to 2016-2023 EPA data



Methane: reported and forecast (Mt CO₂e)

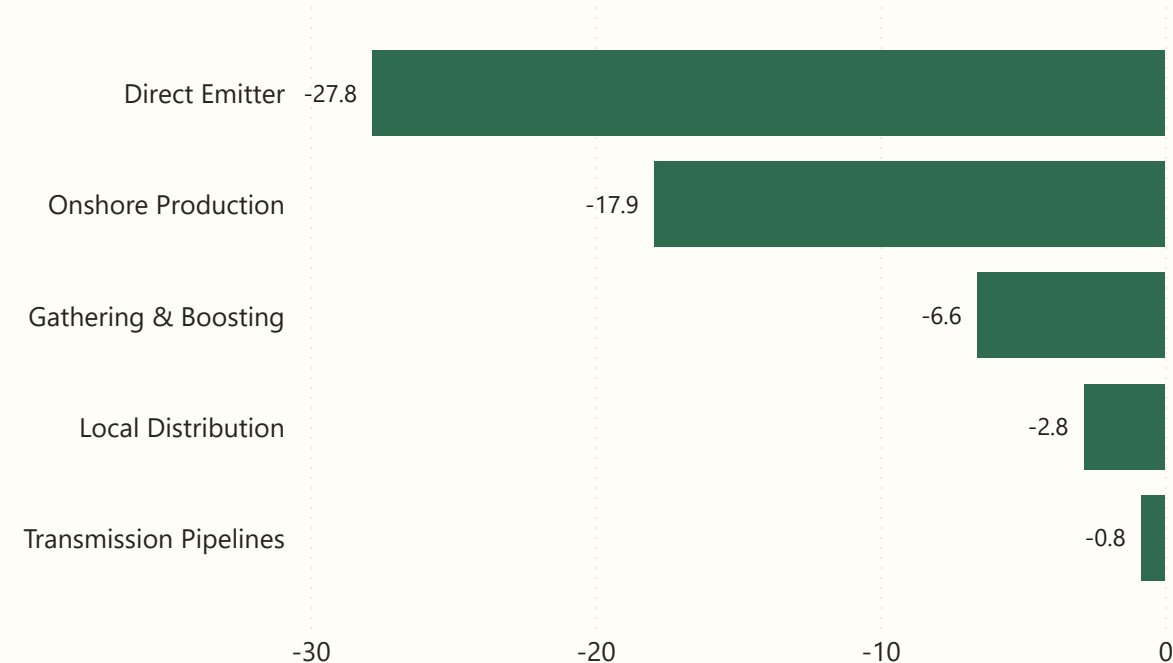
The central forecast meets the pledge by 2030; the top of the 90% range does not

● Reported ● Forecast ● 90% range (high) ● 90% range (low) ● Pledge path



Change by segment, 2023 to selected year (Mt CO₂e)

Direct emitters and onshore production drive most of the decline



Oil and gas methane intensity

Onshore production facilities reporting under EPA Subpart W. Intensity = methane emitted / methane in the gas sold.

Year

All



Methane intensity

0.19%

▼ -17% vs 2022



Methane from production

1,098 kt CH₄

▼ -12% vs 2022



Gas sold by these facilities

1,050 bcm

▲ +6% vs 2022



Methane per producing well

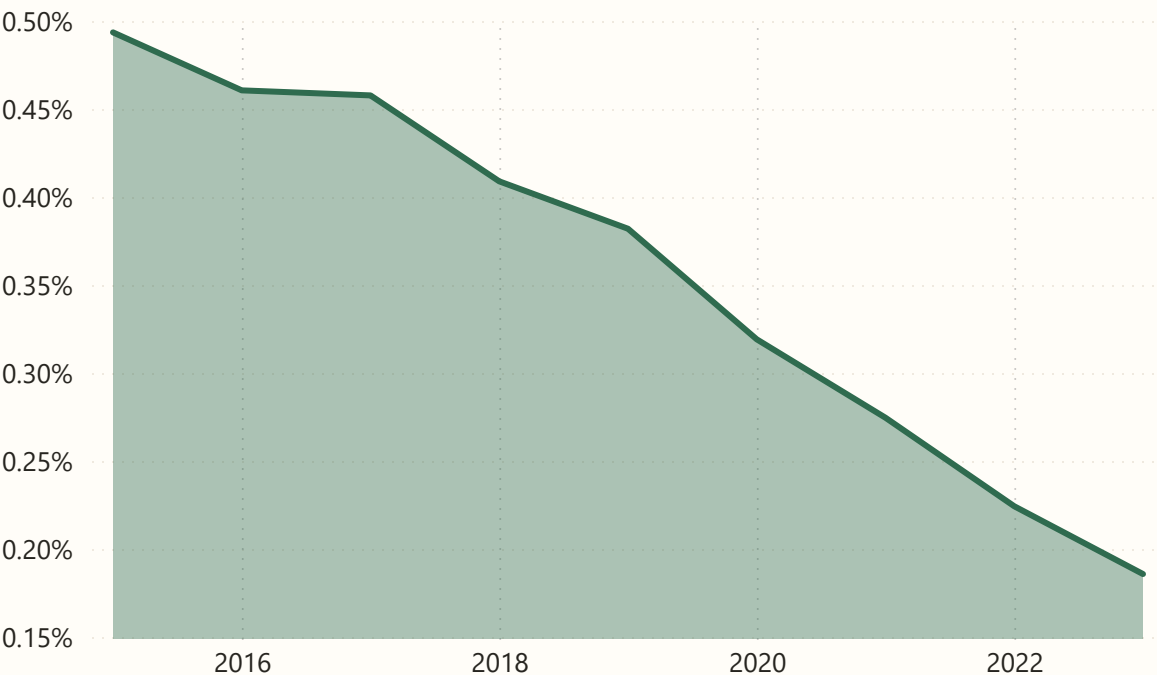
2.29 t CH₄

▼ -13% vs 2022



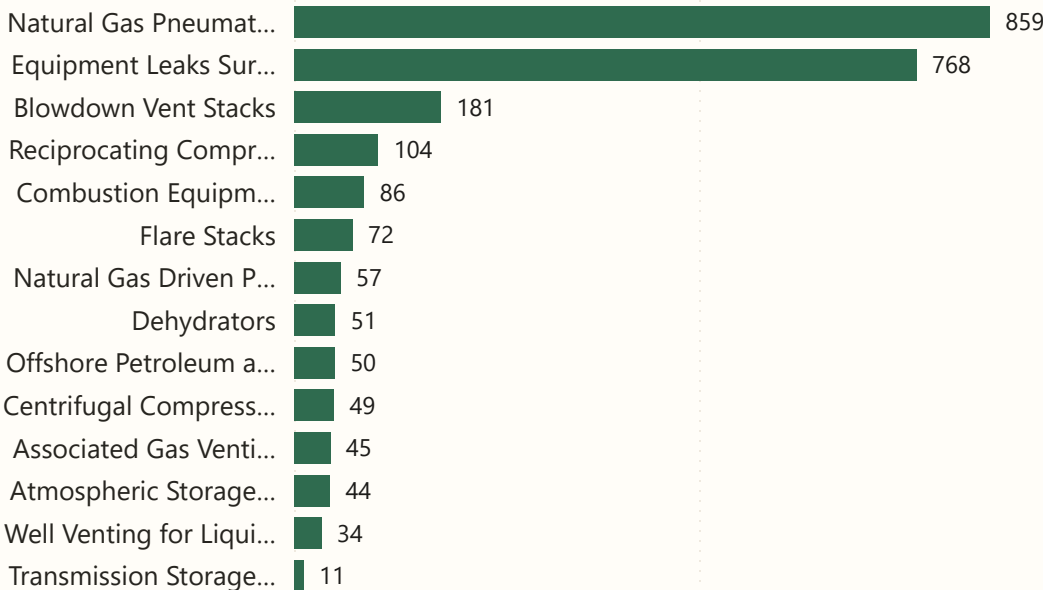
Methane intensity by year

Fell from 0.49% in 2015 to 0.19% in 2023



Oil and gas methane by source, selected year (kt CH₄)

All Subpart W segments, not only production. Pneumatic devices and leaks lead



US exports and EU methane rules

How much US oil and gas trade the EU Methane Regulation touches. Eurostat and EIA data.

Year

All

Exports failing EU rule (%)

All

Penalty or lost value (%)

All

US LNG shipped to the EU

85.9 bcm

▲ +64% vs 2024



EU share of US LNG exports

55.0%

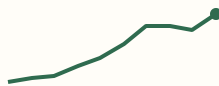
▲ +12.6 pts vs 2024



US share of EU LNG imports

52.8%

▲ +11.7 pts vs 2024



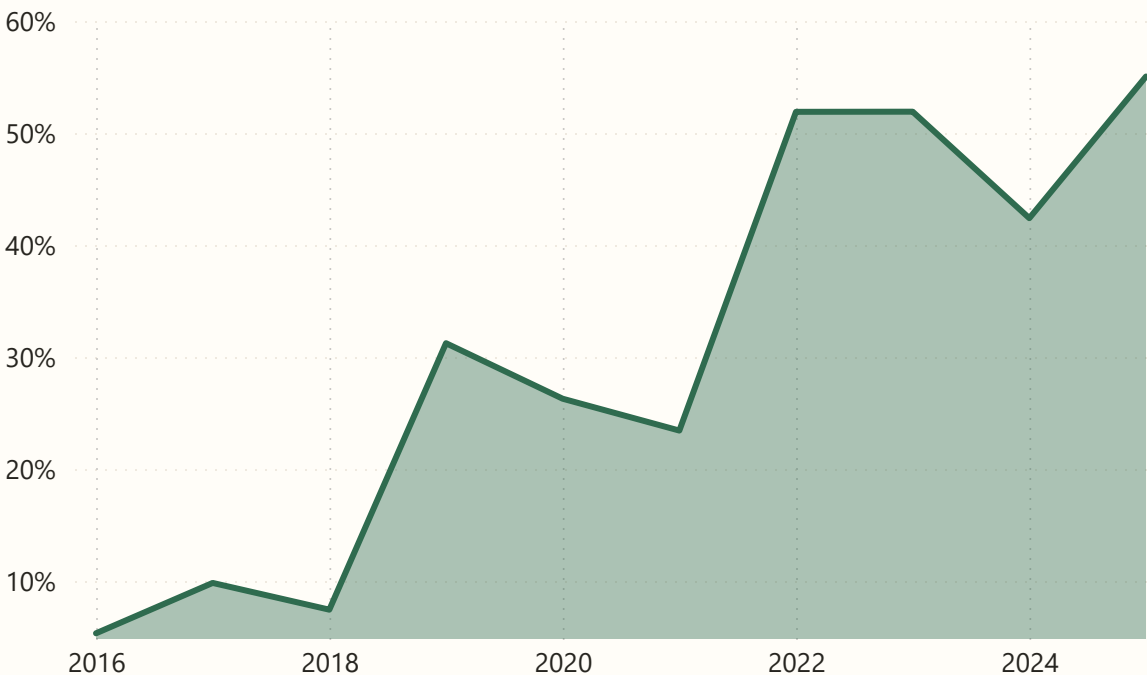
EU import value at risk

\$1.6 bn

50% failing x 5% penalty

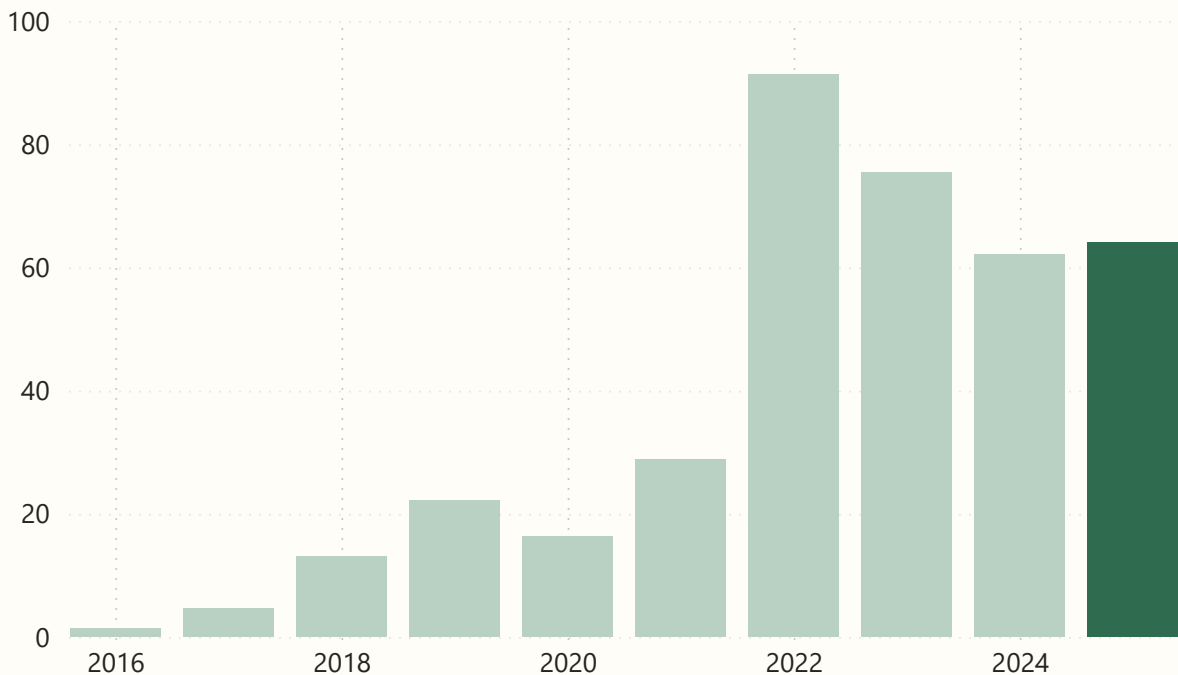
EU share of US LNG exports (%)

From about 10% or less before 2019 to 55% in 2025



EU oil & gas imports from the US (USD bn)

Peaked in 2022 as the EU replaced Russian gas



US methane abatement options

IEA estimates for US oil and gas: how much methane each measure can cut and what it costs net of the gas saved.

Abatement potential

385 Mt CO₂e/yr

All 16 IEA measures, United States

Pays for itself

50 Mt CO₂e/yr

13% of the potential has a negative net cost

Average net cost

\$24.4 per t CO₂e

Weighted by abatement potential

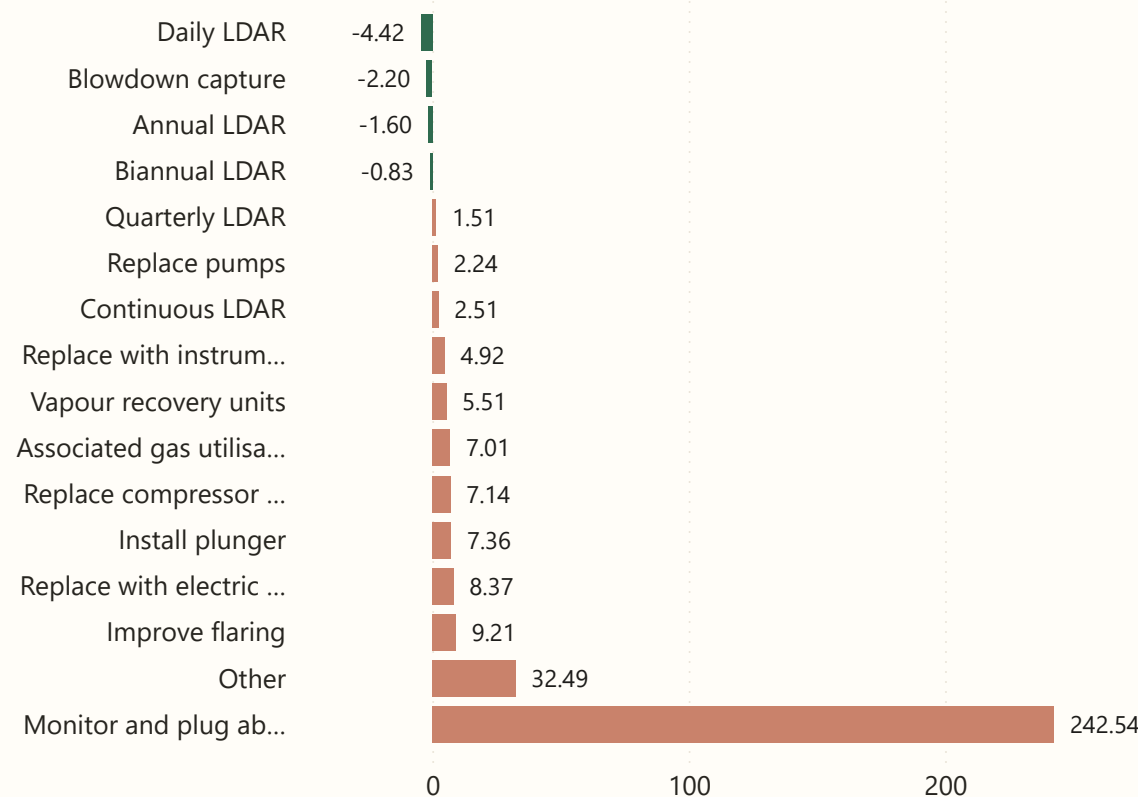
Methane that could be avoided

13.7 Mt CH₄/yr

Equal to the CO₂e figure at GWP 28

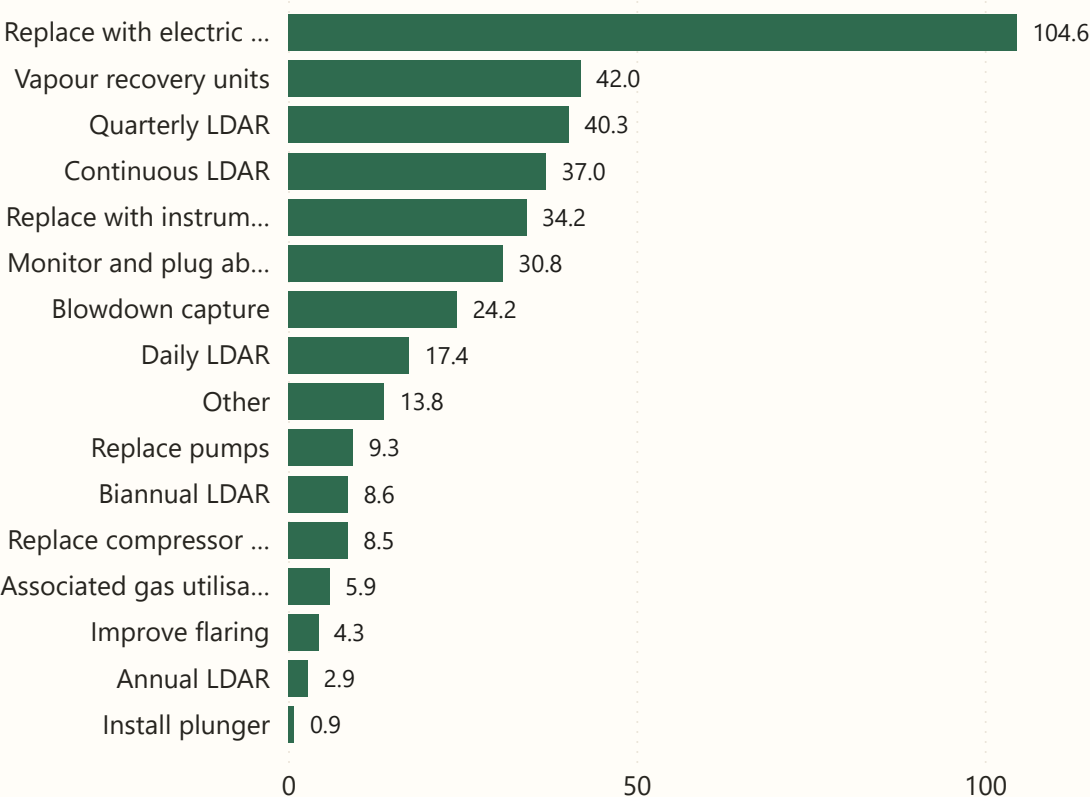
Net cost by measure, cheapest first (USD per t CO₂e)

Leak detection and blowdown capture pay for themselves



Abatement potential by measure (Mt CO₂e per year)

Replacing gas-driven devices with electric motors offers the most



Global oil and gas methane benchmark

IEA Global Methane Tracker, 2025 estimates (includes satellite-detected large leaks).

US oil & gas methane

17.8 Mt CH₄

IEA estimate, 2025

US share of world total

22%

Of all oil and gas methane

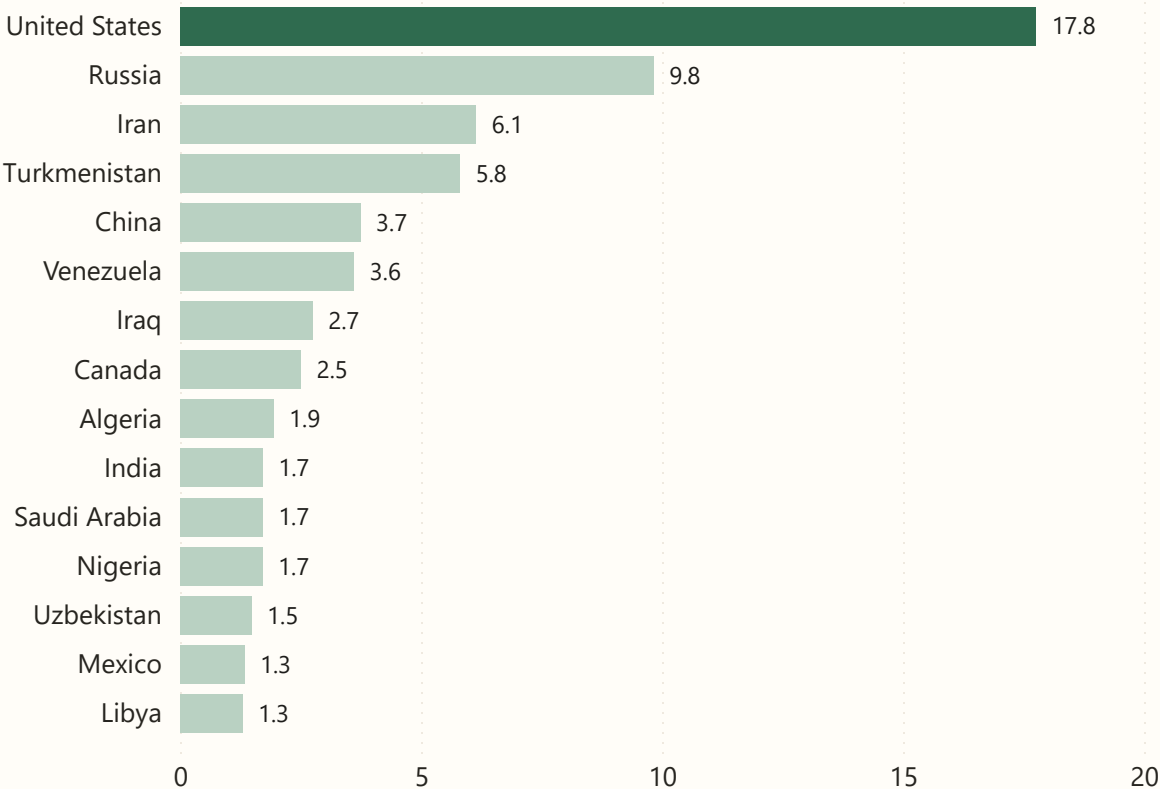
World oil & gas methane

81 Mt CH₄

All countries, 2025

Top 15 countries by oil & gas methane (Mt CH₄)

The US is the largest emitter, ahead of Russia



US oil & gas methane by cause (Mt CH₄)

Venting is about 60% of US oil and gas methane

